Contractile forces sustain and polarize hematopoiesis from stem and progenitor cells.

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Evidence abstract

Self-renewal and differentiation of stem cells depend on asymmetric division and polarized motility processes that in other cell types are modulated by nonmuscle myosin-II (MII) forces and matrix mechanics.

Here, mass spectrometry-calibrated intracellular flow cytometry of human hematopoiesis reveals MIIB to be a major isoform that is strongly polarized in hematopoietic stem cells and progenitors (HSC/Ps) and thereby downregulated in differentiated cells via asymmetric division.

MIIA is constitutive and activated by dephosphorylation during cytokine-triggered differentiation of cells grown on stiff, endosteum-like matrix, but not soft, marrow-like matrix.

In vivo, MIIB is required for generation of blood, while MIIA is required for sustained HSC/P engraftment.

Reversible inhibition of both isoforms in culture with blebbistatin enriches for long-term hematopoietic multilineage reconstituting cells by 5-fold or more as assessed in vivo.

Megakaryocytes also become more polyploid, producing 4-fold more platelets.

MII is thus a multifunctional node in polarized division and niche sensing.

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Evidence checklist

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Source continuity

The canonical evidence above remains the authoritative retrieval target.